Unit Conversions

Prodigy

The ability to convert measurements from one unit to another is critical to anesthetists who must deal in English (US standard) measurements as easily as they deal in the SI (metric) system. Using this section, you will learn how to perform these conversions with one simple rule: multiply the old measurement by a carefully chosen form of the number one. Sound confusing? As you'll see, it's easier than it sounds. The hardest part is knowing the conversion factors to change from one unit to another. Below is a table of conversion factors that you'll use most commonly use in anesthesia-related mathematics.

US Standard to SI Conversions

SYMBOL	WHEN YOU KNOW	MULTIPLY BY	TO FIND	SYMBOL
in	inches	25.4	millimeters	mm
in	inches	2.54	centimeters	cm
ft	feet	0.305	meters	m
yd	yards	0.914	meters	m
fl oz	fluid ounces	29.57	milliliters	mL
gal	gallons	3.785	liters	L
lb	pounds	0.454	kilograms	kg
F	Fahrenheit	(F-32) X 5/9	Celsius	C

SI to US Standard Conversions

SYMBOL	WHEN YOU KNOW	MULTIPLY BY	TO FIND	SYMBOL
mm	millimeters	0.039	inches	in
cm	centimeters	0.39	inches	in
m	meters	3.28	feet	ft
m	meters	1.09	yards	yd
mL	milliliters	0.034	fluid ounces	fl oz
L	liters	0.264	gallons	gal
kg	kilograms	2.2	pounds	Lb
C	Celsius	9/5C + 32	Fahrenheit	F

Because the metric system is based on the number ten, it is much easier to convert units within the metric system than it is between the metric system and the imperial system. Below is a table of common metric prefixes.

Prefix	Abbreviation	Exponent	Number
Giga-	G	10^{9}	1 000 000 000
Mega-	M	10^{6}	1 000 000
kilo-	k	10^{3}	1000
hecto-	h	10^{2}	100
deka-	da	10	10
-	-	-	-
deci-	d	10^{-1}	0.1
centi-	c	10^{-2}	0.01
milli-	m	10^{-3}	0.001
micro-	u (mu)	10^{-6}	0.000 001
nano-	n	10^{-9}	0.000 000 001

For example, converting any value within the metric system to another unit within the metric system is merely a matter of moving the decimal point.

Convert 0.04 kilometers into centimeters.

If we look at the exponent for the kilo prefix, we see that it is 10^3. The exponent for the prefix centi is 10^{-2}. Find the difference between the prefixes.

$3 - (-2) = 3 + 2 = 5$

After determining the difference between the exponents, simply move the decimal point 5 places to the right. We move the decimal to the right if we are shifting to a smaller prefix and to the left if we are shifting to a larger prefix.

Start:	0.04
1:	0.4
2:	4.0
3:	40.0
4:	400.0
5:	4000.0

After moving the decimal point, you can see that there are 4000 centimeters in 0.04 kilometers.

An important point to remember when converting units is that you can multiply anything by 1 and not change its value. This seems obvious, but when performing conversions, this is a crucial component to reasoning through the process.

For example, if you are converting inches into feet, you know that

12 inches = 1 foot

Now, what if you divide both sides by 1 foot?

$$\frac{12 \text{ inches}}{1 \text{ foot}} = \frac{1 \text{ foot}}{1 \text{ foot}}$$

By dividing both sides by the same number and unit (1 foot), we have not changed the relative value of either side. If we reduce the right side of the equation, however, we get:

$$\frac{12 \text{ inches}}{1 \text{ foot}} = 1$$

If 12 inches/1 foot = 1 then according to the associative property, we can multiply a value by 1 or multiply it by (12 inches/1 foot) and get the same answer. This is the property that will allow us to change from one unit to another.

Let's say you need to convert 5.5 feet into inches. You can most likely do this problem in your head quite easily, but trying a simple problem will allow you to see how the process works for more complicated conversions. Start by writing the number you want to convert and multiply it by your conversion factor.

$$5.5 \text{ feet} \times \frac{12 \text{ inches}}{1 \text{ foot}}$$

Which is the same as writing:

$$\frac{5.5 \text{ feet} \times 12 \text{ inches}}{1 \text{ foot}} = 66 \text{ inches}$$

The units for feet cancel each other out leaving 66 inches, your answer in the units you were trying to obtain. Let's try another one that's a little more complicated.

Convert 2.5 liters into gallons.

Looking at our table above, we see that the conversion factor for liters into gallons is 0.264. In other words, there are 0.264 gallons/1 liter. We will set up the conversion as we did in the previous example.

$$2.5 \text{ liters} \times \frac{0.264 \text{ gallons}}{1 \text{ liter}}$$

Rewriting the problem and solving, we get

$$\frac{2.5 \text{ liters} \times 0.264 \text{ gallons}}{1 \text{ liter}} = 0.66 \text{ gallons}$$

Sometimes it is necessary to perform multiple conversions to arrive at an answer. For example:

Convert 4 feet into millimeters.

First, we need to get feet into a unit we can more easily convert into millimeters. We have a conversion factor in our table that converts inches into millimeters, so we'll start by converting feet into inches.

$$\frac{4 \text{ feet}}{1} \times \frac{12 \text{ inches}}{1 \text{ foot}} = 48 \text{ inches}$$

Now that we have the value in inches, we can convert it into millimeters.

$$\frac{48 \text{ inches}}{1} \times \frac{25.4 \text{ mm}}{1 \text{ inch}} = 1219.2 \text{ mm}$$

Unit Conversion Practice Problems

1. How many millimeters in 3.5 inches?
2. How many centimeters in 1 foot 3 inches?
3. How many meters in 215 feet?
4. How many gallons in 6.5 liters?
5. How many milliliters in 5 fluid ounces?
6. Convert 102 fluid ounces into liters.
7. Convert 23 millimeters into inches.
8. How many meters in 1.3 yards?
9. Convert 14 centimeters into inches.
10. How many yards in 7 meters?
11. How many liters are in 10 milliliters?
12. Convert 2 gallons into liters.
13. Convert 280 milliliters into gallons.
14. How many meters are in 14 inches?
15. Convert 50 feet into inches.

16. Convert 10 meters into yards.

17. How many inches are in 2.8 meters?

18. How many liters are in 12 ounces?

19. Convert 2.1 yards into millimeters.

20. Convert 25 meters into yards.

21. How many millimeters are in 21.5 inches?

22. How many fluid ounces are in 5 liters?

23. Convert 113 feet into millimeters.

24. Convert 12.3 milliliters into fluid ounces.

25. Convert 14 yards into millimeters.

Unit Conversion Answers

1.	88.9 mm	6.	3.1 L	11.	0.01 L	16.	10.9 yd	21.	546mm
2.	38.1 cm	7.	0.9 in	12.	7.57 L	17.	110.2 in	22.	170 fl oz
3.	65.6 m	8.	1.2 m	13.	0.07 g	18.	0.36 L	23.	34,442.4 mm
4.	1.7 g	9.	5.5 in	14.	0.36 m	19.	1919.4 mm	24.	0.42 fl oz
5.	149.9 mL	10.	7.6 yd	15.	600 in	20.	27.3 yd	25.	12,796 mm